

SHH gene exploration for the ensembl ICGEB workshop

Assignment Answers

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1. Find and characterize the SHH locus

1) Search for SHH in the human genome in Ensembl

a) What is the **Ensembl Gene ID**?

> [ENSG00000164690.8](#)

b) On which chromosome is SHH located?

> [chromosome 7](#)

c) What are its genomic coordinates in the current human assembly?

> [7:155,793,739-155,818,704](#)

d) On which strand is the gene located?

> [reverse strand](#)

e) What is the approximate genomic span of the SHH locus?

> [7:155,793,739-155,818,704 / 12,484 bp \(could be found in feature explorer \)](#)

2) Identify the genomic assembly being used

a) Is the SHH location the same between **GRCh37 and GRCh38**?

> [GRCh37.p13 : 7:155,586,537-155,611,110](#)

> [GRCh38.p14 : 7:155,793,739-155,818,704](#)

b) If the coordinates differ, explain why.

> The SHH locus is not represented by the same numerical coordinates in GRCh37 and GRCh38. The difference is expected because GRCh37 and GRCh38 are different human reference genome assemblies.

> GRCh38 is a revised reference assembly. During construction of GRCh38, sequence was corrected, added, removed, and/or rearranged in various genomic regions. Consequently, genomic features can acquire different chromosome coordinates even though they represent the same biological locus.

Coordinate changes between assemblies

I noticed that the chromosomal coordinates for a particular gene that I'm looking at have changed since the last time I used your browser. What happened?

A common source of confusion for users arises from mixing up different assemblies. It is very important to be aware of which assembly you are looking at. Within the Genome Browser display, assemblies are labeled by organism and date. To look up the corresponding UCSC database name or NCBI build number, use the [release table](#).

UCSC database labels are of the form hg#, panTro#, etc. The letters designate the organism, e.g. *hg* for human genome or *panTro* for *Pan troglodytes*. The number denotes the UCSC assembly version for that organism. For example, ce1 refers to the first UCSC assembly of the *C. elegans* genome.

The coordinates of your favorite gene in one assembly may not be the same as those in the next release of the assembly unless the gene happens to lie on a completely sequenced and unrevised chromosome. For information on integrating data from one assembly into another, see the [Converting positions between assembly versions](#) section.

source : [UCSC](#)

3) Examine the **Gene** → **Region in detail** view.

a) What genes are located upstream and downstream of SHH?

> Since SHH is on the reverse strand (3'----5'), downstream genes are set on the left and upstream are set on the right ⇒ (downstream-genes - 3'----5' - upstream genes)



source : Figure 3. Representation of downstream and upstream genes based on autonomous TEs' strand. ([Meguerditchian et al.](#))

> you can also check , is the promoter , enhancer is set on the right or on the left.

> downstream genes : INSIG1 (ENSG00000186480.15) - ENS2 (ENSG00000164778.5) - RBM33 (ENSG00000184863.12)

> upstream genes: RNF32 (ENSG00000105982.19) - NOM1 (ENSG00000146909.10)

b) Are there other transcripts or regulatory features surrounding the locus?

> yes there are other 4 transcript + the main shown transcripts

> yes there are an ENSR7_86KNKQ promoter and ENSR7_CKGG2 enhancer

2. Explore SHH transcripts deeply

The Ensembl gene page provides a transcript table containing splice variants, transcript/protein lengths, biotypes and annotations such as MANE, APPRIS and TSL. ([Ensembl Archive documentation](#))

4) How many transcripts are currently associated with **SHH**?

> there are 5 transcripts (see transcript table : [click on me](#))

Get for each SHH transcript, record:

Transcript	Biotype	Transcript length	Protein length	MANE?	APPRIS?	TSL
ENST00000297261.7	protein_coding	4,650	462	yes	yes APPRIS P1	TSL: 1
ENST00000430104.5	protein_coding	677	169	no	no	TSL: 1
ENST00000441114.5	nonsense_mediated_decay	888	126	no	no	TSL: 1

ENST00000435425.1	nonsense_mediated_decay	807	123	no	no	TSL: 1
ENST00000472308.1	protein_coding_CDS_not_defined	504	-	no	no	TSL: 1

5) Do all SHH transcripts encode a protein?

> No the 5th transcript does not encode a protein.

6) Compare the exon structures of the different SHH transcripts. Which exons are shared?

> The second exon is shared by the 4 first transcripts

7) Download the first CDS sequence for the MANE transcript

> [click on the main transcript](#) > [download](#) > [select the CDS option](#) > [download](#)

8) Which SHH transcript is designated as **MANE Select**?

a) What does MANE Select mean?

> [ENST00000297261.7](#)

b) Why might using a standardized transcript be important when reporting a variant?

>MANE select is the most biologically relevant . The transcripts are absolutely identical in both databases, having matching splicing structure, sequence which matches the reference genome, 5' and 3' UTRs and start and end. Check documentation ([click on me](#))

9) Compare the SHH Ensembl transcript with its corresponding **RefSeq transcript**.

c) Are the exon structures identical?

> yes . please refer to the NCBI SSH gene page ([click on me](#)), (GeneID : 6469)

d) Are the transcript and protein sequences identical?

> no, In NCBI there are 10 transcripts ([click on me](#)),

> only 4 transcript are identical (the 1st (NM_000193), the second (NM_001310462), the third (NR_132318), the fourth (NR_132319))

e) What are the corresponding RefSeq identifier of the main Transcript ?

> [NG_007504.2](#)

3. Explore naturally occurring variation in SHH

Ensembl's Variation resources allow you to examine variants associated with genes and transcripts, population frequencies, phenotypes, LD information, and links to external resources.

10) If there, select one the variants you find:

a) deletion > position : 7:155,807,403 / Variant rs1803394776 / Alleles CA C

b) insertion > 7:155,807,409 / Variant rs940900589 / Alleles G GT

c) SNV > 7:155,807,416 / Variant rs1401189945

11) What variant groups are found ?

> Protein altering variants, Splicing variants and Transcript variants

4. Population variation

12) Examine the population frequency information for your SHH variant [rs104894047](#).

> First select variation sidebar and search for rs983483075 then go to [feature explorer](#)

> Allele frequency T (0.0023269)

13) Which populations have the highest and lowest reported allele frequencies?

> 0.001 for South Asian in 1000 Genomes Project Phase 3, check other population

14) Is the variant present in multiple populations or apparently restricted to particular populations?

> no, , check other population

15) What is the source of this variant ? > [dbSNP](#)

5. External database investigation

Ensembl provides cross-references to resources such as UniProt, RefSeq, InterPro and other databases; these links can be accessed from gene/transcript records.

16) What is the source of this variant ? > [dbSNP](#)

17) From the SHH Ensembl page, identify the [external database identifiers](#) for:

- a) HGNC > HGNC:10848
- b) NCBI Gene > 6469
- c) UniProt > Q15465
- d) OMIM > 600725